

Master of Applied Technologies

Course No	COMP8831
Course Name	Machine Learning
Level	8
Credits	15

Student Name	_____
Student ID	_____
Assessment Type	Project Proposal Report
Weighting	10%
Due Date and Time	18/04/2026 23:59
Total Marks	100

Student Declaration

I confirm that:

- This is an original assessment and is entirely my own work.
- The work I am submitting for this assessment is free of plagiarism. I have read and understood the Academic Integrity Procedure. I have also read and understood the Student Disciplinary Statute.
- Where I have used ideas, tables, diagrams etc of other writers, I have acknowledged the source in every case.

Students Signature: _____

Date: _____

1 Assignment Aims

- To propose a machine learning project that integrates the skills and knowledge gained through the course.
- To achieve greater awareness of the latest artificial intelligence tools and techniques for prediction, recognition, optimisation, and decision making.
- To understand the principles, advantages, limitations and possible applications of machine learning algorithms.

2 Assignment Task

In this assignment, you will write a proposal for a machine learning project that you will later implement as your final project. Think of this proposal as the blueprint for your final project. You are defining the problem, choosing your methods, and planning your approach now so that you have a clear direction when you begin the implementation. A well-written proposal will make your final project significantly easier to execute, so invest the time to plan thoroughly.

You are to search, select, and propose a machine learning project in a written report. Your proposal should be between **1500 and 2500 words** and should NOT exceed **six pages**. The word count and page limit both exclude the title page and references. Use font size 11–12, Times family, single spacing.

The proposal weighs **10 percent** of your final mark, so it needs to be written carefully.

Each group is required to submit only one proposal. All projects (whether individual or group) must train and compare multiple models. In a two-member group, at least two different learning methods need to be discussed for the implementation.

Your proposed project should fit supervised, unsupervised, or reinforcement learning domains.

Important Requirements

- Choose a problem that is substantive enough to serve as a capstone ML project. Make sure you can clearly justify its complexity and real-world relevance.
- **Model Requirements:** Your primary solution must use modern, state-of-the-art machine-learning techniques (e.g., neural networks, advanced ensemble models, deep learning architectures, transformer variants, graph neural networks, diffusion models, etc.).
- **Baseline Comparisons:** Classical methods, like logistic or linear regression, SVMs, random forests, and decision trees, are welcome only as baseline benchmarks. They must not be your proposed method.

3 Proposal Structure

Your proposal must include the following sections in this order. Approximate word counts are provided as guidance.

3.1 Title Page

Include the project title, full name(s) of group member(s), student ID(s), course code, and date.

3.2 Table of Contents

List all sections and subsections with page numbers.

3.3 Abstract (150–250 words)

Provide a concise summary of your project. It should cover:

- The problem you intend to solve.
- The machine learning approach you propose.
- The expected outcome.

3.4 Problem Statement (200–400 words)

Define the problem your project addresses:

- What is the problem and why does it matter?
- What is the real-world significance or application?
- Why is machine learning a suitable approach for this problem?

3.5 Background / Brief Literature Review (500–800 words)

Demonstrate awareness of existing work in your chosen area:

- Review 5–8 key existing works (papers, systems, or methods) related to your problem.
- Summarise relevant approaches that have been applied to similar problems.
- Show how existing work informs your choice of methods and dataset.

3.6 Project Methodology (500–1000 words)

Describe the technical details of your proposed implementation:

Preferred Learning Methods

Identify the specific ML algorithms or architectures you plan to use. Explain how they work at a high level.

Required Datasets

Specify the datasets you will use for training and testing. Include the source, approximate size, and key features. If using multiple datasets, explain the role of each.

Method Justification

Explain why your chosen methods are appropriate for the problem and why you selected these specific models for comparison.

Tools and Languages

State your preferred programming language, libraries, and frameworks.

3.7 Expected Outcomes / Deliverables (150–250 words)

Describe what your completed project will produce:

- A cleaned and processed dataset ready for modelling.
- Multiple trained models applied to the same dataset.
- Evaluation of each model using appropriate performance metrics.
- A comprehensive comparison of model results.
- A recommended final model with justification for why it is the best fit for the problem.

3.8 References

List all sources cited in your proposal using **IEEE referencing style**. You must include a minimum of **8 quality sources** (peer-reviewed papers, conference proceedings, books, or reputable technical resources).

Examples of IEEE formatting can be found at:

<http://libguides.murdoch.edu.au/IEEE/sample>

4 Marking Criteria (100 Marks)

Criteria	Marks	What We Assess
Abstract	5	Clear summary of problem, approach, and expected outcome
Problem Statement	10	Clearly defined problem, real-world relevance, justification for ML
Background / Literature Review	15	Awareness of existing work, relevant sources reviewed
Learning Methods	15	Appropriate modern methods selected, technical understanding demonstrated
Dataset Plan	10	Suitable data identified, source cited, size and features described
Method Justification	15	Why chosen methods fit the problem, rationale for model selection
Expected Outcomes	10	Clear deliverables, multiple models compared, final model proposed
Writing Quality & Structure	10	Academic tone, logical flow, formatting, grammar
References	10	IEEE style, minimum 8 quality sources, properly cited in text
Total	100	

5 Report Format

- Use MS Word or LaTeX for preparing your report.
- Number all sections and subsections.
- Number all pages.
- Ensure the table of contents matches section numbers and page numbers.
- Put captions on all figures, tables, and diagrams.
- Place page breaks appropriately (not immediately after headings or in the middle of a figure/table).
- Cite all work from other sources in the text and list them in your references.
- Most of your report should be written in your own words. Each quotation, table, or diagram from another source must be accompanied by your own discussion of what it shows, why you included it, and why it is relevant.

6 Submission

Upload your proposal into Moodle through the Turnitin Assignment Submission module (under Assessments, “Project Proposals Submission Link”) before the due date. The Turnitin similarity score, excluding references, should be less than **10%**.

Each proposal is reviewed by the lecturer, and individual feedback will be provided. Once the proposal gets approved, the group can start the implementation of the second assignment accordingly (the instructions will be provided separately).

Policies

Academic Integrity

Marks will be deducted if you fail to draw upon appropriate professional and academic literature or if the assignment is not written and presented in a professional manner. When choosing sources, check that they have passed some form of quality assurance (books, journal articles, and conference articles are a good choice). If it is difficult to identify the author, year, or title of an article then it has not been subject to quality assurance.

Plagiarism will be reported to the Programme Authorities who may award zero marks or refer the matter to the Discipline Committee, which has powers of suspension/exclusion.

Late Submission of Assignments

Assignments submitted after the due date and time without having received an extension through Affected Performance Consideration (APC) will be penalised according to the following:

- **10%** of marks deducted if submitted within 24hrs of the deadline.
- **20%** of marks deducted if submitted after 24hrs and up to 48hrs of the deadline.
- **No grade** will be awarded for an assignment that is submitted later than 48hrs after the deadline.

Assignments submitted in more than 48 hours late will not be marked unless Affected Performance Consideration (APC) apply. So, it is better to submit an incomplete assignment on time.

Affected Performance Consideration

A student, who due to circumstances beyond his or her control, misses a test, final exam or an assignment deadline or considers his or her performance in a test, final exam or an assignment to have been adversely affected, should complete the Affected Performance Consideration (APC) form available from Student Central or online:

<https://www.unitec.ac.nz/current-students/study-support/affected-performance-consideration>

Assistance to Other Students

Students themselves can be an excellent resource to assist the learning of fellow students, but there are issues that arise in assessments that relate to the type and amount of assistance given by students to other students. It is important to recognise what types of assistance are beneficial to another's learning and also what types of assistance are unacceptable in an assessment.

Beneficial Assistance:

- Study Groups.
- Discussion.
- Sharing reading material.
- Testing another student's programming work using the executable code and giving them the results of that testing.

Unacceptable Assistance:

- Working together on one copy of the assessment and submitting it as own work.
- Giving another student your work.
- Copying someone else's work. This includes work done by someone not on the course.
- Changing or correcting another student's work.
- Copying from books, Internet etc. and submitting it as own work.

Do you want to do the best that you can do on this assignment and improve your grades? You could:

- Talk it over with your lecturer
- Visit Student Success and Achievement for learning advice and support
- Visit the Centre for Pacific Development and Support
- Visit the Centre for Maori Development and Support